Generated by Scopus AI, Sat Oct 10 2026

Smart cybersecurity tool for students

Smart Cybersecurity Tools for Student Protection: Real-Time Detection and Alerting of Online Threats

Quick Reference

Key Findings Table

Theme/Aspect	Key Insights	Supporting Citations
Hybrid AI Architectures	Hybrid models (ML + LLM) outperform standalone approaches in phishing detection, balancing accuracy and efficiency.	1 2 3
User-Centric Design & Awareness	Gamified, interactive, and explainable tools significantly improve student engagement and phishing resilience.	4 5 6
Privacy Preservation	Federated learning, homomorphic encryption, and SMPC are essential for student data confidentiality.	7 8 9
Adversarial Robustness	Adaptive thresholding, adversarial training, and continuous updates are critical against evolving threats.	10 11 12
Explainability & Trust	XAI methods (SHAP, LIME, attention) increase transparency, trust, and compliance in student tools.	13 14 15
Educational Integration	AI-powered platforms that personalize learning and integrate real-time feedback enhance phishing awareness.	5 16 17
Real-Time Detection Gaps	High false positive/negative rates, zero-day attacks, and adversarial threats remain key challenges.	18 19 20

Theme/Aspect	Key Insights	Supporting Citations
Demographic Moderators	Gender, age, and neurodiversity influence awareness, susceptibility, and tool effectiveness.	21 22 23

Direct Answer

The research indicates that developing a smart cybersecurity tool for students should involve a hybrid AI approach that combines traditional machine learning with LLM-driven semantic analysis, integrated within a user-centric design framework that emphasizes interactive, gamified training and explainable alert mechanisms. Privacy-preserving techniques such as federated learning and homomorphic encryption are essential to protect student data, while continuous model updates and robust adversarial defenses will ensure adaptability against evolving phishing threats.

Study Scope

- **Time Period:** 2019–2024 (focus on recent advances)
- **Disciplines:** Cybersecurity, AI/ML, Education Technology, Human-Computer Interaction, Privacy Engineering
- **Methods:** Meta-analysis, empirical studies, case studies, technical evaluations, behavioral research, and user studies

Assumptions & Limitations

- Most hybrid AI models are evaluated in controlled or pilot deployments; large-scale, real-world educational settings remain underexplored.
- Neurodivergent student populations are underrepresented in empirical phishing research.
- Privacy-preserving techniques may introduce computational overhead, potentially impacting real-time performance on low-resource devices.
- Effectiveness of gamified and explainable interventions may vary by cultural, demographic, and institutional context.

Suggested Further Research

- Longitudinal studies on the deployment and efficacy of hybrid AI tools in diverse educational environments.
- Empirical research on phishing susceptibility and tool effectiveness among neurodivergent students.
- Optimization of privacy-preserving AI for real-time, resource-constrained student devices.
- Development of adaptive, context-aware educational interventions integrated with detection tools.

1. Introduction

Context and Motivation

The digital transformation of education has exposed students to a rapidly evolving landscape of online threats, including phishing attacks, suspicious links, fake websites, and unsafe emails. Students are particularly vulnerable due to their high engagement with digital platforms, limited cybersecurity experience, and the increasing sophistication of social engineering tactics targeting educational communities. Despite the proliferation of anti-phishing tools and awareness campaigns, gaps persist in both technical protection and user awareness, necessitating the development of smart, proactive cybersecurity tools tailored to student needs. These tools must not only detect and alert users to threats before interaction but also foster resilience through education, transparency, and privacy protection [18] [24] [25].

2. Landscape of Smart Cybersecurity Tools for Students

Overview of Existing Tools

A variety of smart cybersecurity tools have emerged to address online threats in educational contexts:

- **SafeSurf-AI:** A hybrid browser-based tool integrating rule-based heuristics, machine learning, and explainable AI for real-time URL inspection and user awareness simulation, achieving high accuracy and responsiveness [24].
- **CyberShield:** A browser plugin leveraging ML and DL models (XGBoost, Random Forest, neural networks) for real-time phishing detection with identification rates above 99% [26].
- **SecureNet Lite:** An AI-powered platform combining phishing link classification, password breach detection, and intelligent login monitoring, optimized for student usability [27].

- **LLM-based Browser Extensions:** Extensions utilizing large language models for advanced text analysis and phishing detection, offering accessible, user-friendly protection [28].
- **Interactive Education Platforms:** Tools like KnowBe4, PhishingBox, and Gophish, as well as gamified simulations, enhance awareness and engagement, improving assessment scores by up to 25% [29] [30] [31].

These tools typically combine automated detection with educational interventions, but student reliance and trust remain limited, highlighting the need for integrated, user-centric solutions [25] [32] [33] [34].

Detection Techniques and Alert Mechanisms

- **Technical Approaches:** ML, DL, NLP, and hybrid models analyze email content, website structure, URLs, and behavioral patterns. Feature extraction techniques (TF-IDF, CNN, LSTM, ensemble methods) improve generalization and reduce false positives [35] [36].
- **Hybrid Models:** Integration of ML, NLP, blockchain, federated learning, adversarial detection, and explainable AI enhances effectiveness and trust [35] [37] [38].
- **Alert Mechanisms:** Browser extensions, real-time notifications, gamified quizzes, and simulated phishing campaigns are common. Gamified interventions and interactive modules significantly improve recognition accuracy and motivation [4] [28] [39] [40].
- **OSINT Integration:** Real-time threat monitoring systems leverage open-source intelligence from social media, combined with NLP and ensemble classifiers, to enhance incident response [41] [42].

Adoption and Usability Among Student Demographics

- **Adoption Rates:** AI-powered tools are widely used (70–80% adoption), especially for academic assistance, but trust in outputs is low (<25%) [43] [44] [45].
- **Demographic Influences:** Usage varies by gender, age, academic level, and field of study. Male and younger students show higher usage; females demonstrate more prudent privacy behaviors [21] [44] [46] [47].
- **Barriers:** Lack of formal training, overconfidence, and ethical concerns persist. Students rely on self-directed learning, risking overconfidence without critical evaluation skills [48] [49] [50].
- **Personas:** Adoption personas range from cautious to enthusiastic, shaped by trust, perceived usefulness, and ethical comfort, underscoring the need for tailored AI literacy initiatives [51].

Synthesis: The current landscape demonstrates technical maturity in detection but highlights persistent gaps in user trust, engagement, and demographic inclusivity. Integrating detection with adaptive, user-centric education is essential for effective student protection.

3. Advanced Detection Architectures and Real-Time Threat Monitoring

Hybrid LLM-ML Cascade Architectures

- **Cascade Models:** Hybrid architectures (e.g., XGBoost + LLM) selectively invoke LLMs for uncertain cases, achieving high accuracy (F1=0.9662) and cost-efficiency suitable for browser extensions [1].
- **Multi-Model Ensembles:** Integrating Random Forest, CNN-LSTM, TinyBERT, and Autoencoder models achieves up to 98.6% accuracy, robust against obfuscated and zero-day attacks [52].
- **PhishCascade:** A confidence-gated, multi-agent framework using adaptive URL filtering and selective LLM analysis achieves F1=0.9811 with reduced latency [2].
- **Practical Deployment:** Tools like PhishGuard and SafeClick demonstrate real-time, scalable detection with explainable alerts, suitable for student devices [53] [54].

Retrieval-Augmented Generation (RAG) for Personalized Alerts

- **Personalization:** RAG integrates retrieval and generation modules, enhancing factual accuracy and reducing hallucinations by retrieving relevant external knowledge, critical for precise, context-aware student alerts [55] [56].
- **Modular Optimization:** Hierarchical indexing, query expansion, and re-ranking reduce irrelevant retrievals and false positives [56] [57].
- **Security:** Robust defenses (data filtering, adversarial training) are essential to maintain integrity and reduce false positives in student-focused tools [55] [58].

Computational and Energy Trade-offs

- **Standalone LLMs:** High false positives, longer preprocessing, and increased operational burden on student devices [59] [60].
- **Hybrid Models:** RAG-based and cascade approaches reduce false positives by up to 66.7%, lower computational overhead, and maintain high accuracy (F1=0.9703) [59] [60].

- **Cost-Optimized Systems:** Multi-agent LLM systems (e.g., CLASP) maintain high accuracy with low per-instance processing time and manageable API costs [61].
- **Lightweight Models:** Quantized LLMs and transformer-based URL detectors offer energy-efficient deployment on student devices [62] [63].

Semantic Analysis for AI-Generated Phishing Content

- **Transformer Models:** BERT, RoBERTa, and DistilBERT enable high-accuracy detection of AI-generated phishing, with tri-class classification distinguishing legitimate, human-written, and AI-generated phishing [64] [65].
- **Knowledge Distillation:** Transfers capabilities from large models to smaller, efficient ones (e.g., TinyBERT), supporting real-time, scalable detection [66].
- **Multi-Modal Approaches:** Combining semantic, visual, and contextual cues enhances detection and explainability [67] [68].

Synthesis: State-of-the-art architectures balance detection accuracy, latency, and resource constraints, with hybrid and RAG-based models offering scalable, personalized, and explainable solutions for student-focused cybersecurity tools.

4. Student User Behavior, Awareness, and Psychological Factors

Awareness and Response to Online Threats

- **Awareness Levels:** Students exhibit moderate awareness but insufficient reliance on anti-phishing tools; integrated education and advanced tools enhance resilience [25] [69].
- **Behavioral Patterns:** Students manage risks through routines, optimistic bias, and self-efficacy but may experience learned helplessness [70].
- **Knowledge Gaps:** Many students do not fully recognize phishing dangers, underscoring the need for improved detection systems and education [32] [71].
- **Demographic Variability:** Male and older students tend to have higher knowledge and lower anxiety; neurodivergent students are underrepresented in research [72] [73] [74].

Psychological Influences on Tool Trust and Reliance

- **Key Factors:** Perceived threat severity and self-efficacy are strong predictors of security behavior; trust is influenced by tool transparency and understanding [15] [75].
- **Emotional Traits:** Anxiety and risk tolerance significantly affect susceptibility; impulsivity, trust, and cognitive engagement also play roles [76] [77].
- **Overconfidence:** Students often overestimate their detection abilities, highlighting the need for interactive, confidence-building interventions [31].

Demographic Moderators: Gender, Age, and Neurodiversity

- **Gender:** Males emphasize self-exploration; females prefer formal education and demonstrate prudent privacy behaviors [21] [47].
- **Age:** Older students are more cautious but not necessarily more accurate; younger students are more vulnerable [22].
- **Neurodiversity:** Neurodivergent students face higher cognitive load and less peer support, but direct comparisons with neurotypical peers are limited [23] [78].

Gamified Cybersecurity Awareness Interventions

- **Effectiveness:** Gamified simulations, quizzes, and serious games significantly improve comprehension, motivation, and retention compared to traditional methods [4] [5].
- **Design Principles:** Grounding in behavioral theories (e.g., Protection Motivation Theory), adaptive feedback, and real-world scenarios enhance impact [75] [79].
- **Differential Impact:** Effectiveness varies by gender, age, and field of study, necessitating tailored designs [80] [81].

Synthesis: Student behavior and psychological factors are critical to tool adoption and effectiveness. Gamified, personalized, and explainable interventions, sensitive to demographic diversity, are essential for building resilience.

5. Technological Gaps and Challenges in Real-Time Detection

Challenges in Real-Time Detection and Alerting

- **Zero-Day Attacks:** Evolving tactics, obfuscated URLs, and AI-generated content challenge traditional detection [18] [19] [20].
- **False Positives/Negatives:** High rates stem from limited feature sets, static patterns, and poor generalization [82] [83].

- **Integration Barriers:** Student-focused tools often lack tailored education and awareness components; privacy concerns hinder incident reporting [32] [84].
- **Noisy Data:** Filtering and categorizing threats from social media and diverse sources remains a technical hurdle [41].

False Positives and Negatives: Causes and Solutions

- **Feature Expansion:** Incorporating lexical, host-based, content-based, and n-gram features improves accuracy [82].
- **Ensemble Learning:** Combining multiple classifiers and feature types enhances robustness [85] [86].
- **Adaptive Thresholding:** Online model updates and adaptive learning maintain low error rates against evolving tactics [11] [87].
- **Visual Similarity:** Integrating visual and textual features reduces misclassification, especially for visually deceptive phishing [88].

Adversarial Attacks and AI-Generated Misinformation

- **Model Vulnerabilities:** ML/DL models degrade under adversarial shifts; specialized models (e.g., URLBERT) offer improved robustness [89].
- **Defensive Strategies:** Adversarial training, causal modeling, and multi-modal approaches enhance resilience [90] [91].
- **Dataset Augmentation:** Synthetic adversarial examples improve robustness but require careful integration [92] [93].

Filtering Noisy Social Media Data for Accurate Alerts

- **Deep Learning:** CNNs and transformer-based NLP models (BERT) effectively filter noise and extract context [94] [95].
- **Multimodal Fusion:** Combining textual, metadata, and visual features with lightweight models supports real-time detection on student devices [96].
- **Explainable AI:** Enhances transparency and interpretability, critical for trust and regulatory compliance [97].

Synthesis: Addressing technological gaps requires advanced feature extraction, adaptive learning, robust adversarial defenses, and integration of explainable, privacy-preserving AI.

6. AI Integration, Privacy, and Usability Considerations

AI and ML Integration for Enhanced Detection

- **Techniques:** Anomaly detection, malware analysis, and real-time intrusion detection improve speed and precision [98] [99] [100].
- **Ethical Alignment:** Governance frameworks ensure fairness, transparency, and privacy [98] [99] [101].
- **Behavioral Analytics:** Explainable AI and user-centric design improve transparency and usability [100] [102].

Privacy-Preserving AI Techniques

- **Federated Learning:** Enables collaborative model training without sharing raw data, maintaining confidentiality and high accuracy [7] [103].
- **Homomorphic Encryption & SMPC:** Allow encrypted data analysis, balancing privacy with computational efficiency [8].
- **Data Augmentation:** LLM-driven synthetic data generation enhances robustness without exposing sensitive data [104] [105].

Explainable AI for Trust and Transparency

- **XAI Methods:** SHAP, LIME, and attention visualization clarify model decisions, increasing trust and compliance [106] [107].
- **Operational Impact:** XAI reduces triage time and increases analyst confidence, supporting effective student engagement [108].
- **Regulatory Compliance:** XAI supports GDPR and AI Act requirements for auditable, explainable decisions [109].

Ethical Governance Frameworks in Educational Contexts

- **Responsible AI:** Frameworks emphasize ethical design, risk-based audits, and modular governance for compliance and trust [110] [111] [112].
- **Transparency Analytics:** Structured risk registers and policy mapping support adaptive, transparent governance [113].
- **Curricular Integration:** Ethics and professional skills development are essential for preparing future cybersecurity professionals [114].

Synthesis: Integrating advanced AI with privacy-preserving, explainable, and ethically governed frameworks is essential for effective, trustworthy, and compliant student cybersecurity tools.

7. Future Directions: Adaptive Learning, Federated AI, and Robustness

AI-Powered Educational Platforms for Phishing Awareness

- **Personalization:** Diagnostic assessments and adaptive lesson sequencing tailor learning to individual needs [16].
- **Real-Time Feedback:** Continuous analysis and sentiment detection dynamically adjust content delivery [115] [116].
- **Gamification:** Enhances engagement, motivation, and retention, supporting practical phishing response skills [117].

Federated Learning and LLMs for Privacy and Accuracy

- **Frameworks:** Hybrid Transformer-CNN and multi-branch federated transformer models achieve high accuracy (>95%) with strong privacy preservation [118] [119] .
- **Parameter-Efficient Fine-Tuning:** Reduces communication overhead, enabling scalable, privacy-preserving adaptation [120] [121].
- **Explainability:** XAI techniques embedded in federated pipelines enhance transparency without compromising privacy [122].

Privacy-Preserving Data Augmentation with LLMs

- **Synthetic Data:** LLM-generated data improves robustness and reduces false positives without exposing sensitive information [105] [123].
- **Privacy Techniques:** Anonymization, prompt tuning, and differential privacy mitigate exposure risks [124] [125].
- **Local Processing:** Retrieval-augmented generation systems process data within secure environments, minimizing leakage [126].

Synthesis: Future research should focus on adaptive, privacy-preserving, and explainable AI platforms that personalize education, enhance detection, and ensure robust protection for diverse student populations.

8. Conclusion

Summary and Recommendations

This review synthesizes evidence that effective student-focused cybersecurity tools must:

- **Employ Hybrid AI Architectures:** Combine traditional ML with LLM-driven semantic analysis for robust, adaptable, and efficient phishing detection [1] [2] [3].
- **Integrate User-Centric Design:** Use gamified, interactive, and explainable mechanisms to enhance awareness, trust, and engagement [4] [5] [6].
- **Prioritize Privacy:** Implement federated learning, homomorphic encryption, and SMPC to protect student data [7] [8] [9].
- **Ensure Adversarial Robustness:** Continuously update models, employ adaptive thresholding, and integrate adversarial defenses [10] [11] [12].
- **Foster Explainability and Trust:** Leverage XAI to clarify decisions, support compliance, and build user confidence [13] [14] [15].
- **Address Demographic Diversity:** Tailor interventions to gender, age, and neurodiversity for inclusive effectiveness [21] [22] [23].
- **Bridge Technical and Educational Gaps:** Combine detection with adaptive, context-aware educational interventions for holistic protection [5] [16] [17].

Recommendations:

- Developers should prioritize hybrid, privacy-preserving, and explainable architectures.
- Educators and policymakers must integrate adaptive, gamified cybersecurity education into curricula.
- Future research should address underexplored populations (e.g., neurodivergent students) and optimize real-time, resource-efficient deployment.

By bridging technical innovation with user-centric, ethical, and educational strategies, the next generation of smart cybersecurity tools can empower students to navigate the digital world safely and confidently.

References

1. A Hybrid ML-LLM Cascade System for Cost-Efficient Phishing Webpage Detection Ozaslan, Y., Fatih Adak, M. Proceedings - 2026 10th International Young Engineers Forum on Electrical and Computer Engineering, YEF-ECE 2026, 2026 https://www.scopus.com/pages/publications/105047081380?origin=scopusAI

2. PhishCascade: Confidence-Gated Multi-agent Phishing Detection Li, H., Lin, Y. Lecture Notes in Computer Science, 2026 https://www.scopus.com/pages/publications/105047061484?origin=scopusAI

3. Phishing Email Detection Using Large Language Models: A Comparative Analysis of Standalone and Retrieval Augmented Generation Models Chandaeng, T., Kosolsombat, S., Ratanavilisagul, C. Proceedings - 2025 10th International Conference on Computational Intelligence and Applications, ICCIA 2025, 2025 https://www.scopus.com/pages/publications/105032923444?origin=scopusAI

4. PhishWise: Enhancing Students Understanding of Social Engineering Awareness in Phishing Through Gamification Wijayanti, R.A., Qomariasih, N., Siddiq, S., (...), Natalia, M.C. Proceedings - International Conference on Education and Technology, ICET, 2024 https://www.scopus.com/pages/publications/85214709510?origin=scopusAI

5. Gamification Tools for Social Engineering Attack Awareness: An AI-Driven Approach Bhimavarapu, U. Gamification Learning Framework for Cybersecurity Education, 2025 https://www.scopus.com/pages/publications/105017950692?origin=scopusAI

6. Generalized Explainable AI Framework for Phishing Detection on Heterogeneous Textual Data Mansour, L., Hilal, N., Abbas, N., Kadry, S. Computer and Decision Making, 2026 https://www.scopus.com/pages/publications/105045082155?origin=scopusAI

7. Privacy-Preserving Federated Learning for Phishing Detection Elkhawas, A.I., Chen, T.M., Gashi, I. IEEE Technology and Society Magazine, 2025 https://www.scopus.com/pages/publications/105004950307?origin=scopusAI

8. Privacy-preserving over encrypted data for web page phishing detection Bermani, A.K., Al-Salih, A.M., Jabir, H.A., Manaa, M.E. Journal of Discrete Mathematical Sciences and Cryptography, 2025 https://www.scopus.com/pages/publications/105011935772?origin=scopusAI

9. A Spam Detection Model With Federated Learning and Large Language Model Zhao, J., Wen, B., Yang, J., Zhou, S. Communications in Computer and Information Science, 2026 https://www.scopus.com/pages/publications/105028244310?origin=scopusAI

10. Combatting AI-Generated Phishing Attacks with Adaptive Threat Intelligence Balsem, Z.A., Wbaid, S., Pavithra, A., (...), Amonov, Sh.N. Adaptive Technologies for Sustainable Growth, 2026 https://www.scopus.com/pages/publications/105033323726?origin=scopusAI

11. Real Time Attack Detection using Machine Learning Techniques Chirayil, D., Singh, A., Kumar, A., Lalwani, S. Proceedings of 2025 International Conference on Emerging Trends in Industry 4.0 Technologies, ICETI4T 2025, 2025 https://www.scopus.com/pages/publications/105016399381?origin=scopusAI

12. The anatomy of a resilient phishing detector: an in-depth analysis of a transformer's interpretability, causality, and robustness D., V., Dayalan, G., Bhandarkar, A. Frontiers in Computer Science, 2026 https://www.scopus.com/pages/publications/105050561905?origin=scopusAI

13. Improving cybersecurity through explainable artificial intelligence: a systematic literature review Oriaro, S., Mishra, S. Issues in Information Systems, 2025 https://www.scopus.com/pages/publications/105018691220?origin=scopusAI

14. Bridging the gap: explainable AI in threat detection and cybersecurity Naeem, M.R., Farhan, M. Explainable Artificial Intelligence (XAI) for Next Generation Cybersecurity: Concepts, Challenges and Applications, 2025 https://www.scopus.com/pages/publications/105026740357?origin=scopusAI

15. It's not just about accuracy: An investigation of the human factors in users' reliance on anti-phishing tools Schuetz, S.W., Steelman, Z.R., Syler, R.A. Decision Support Systems, 2022 https://www.scopus.com/pages/publications/85135842165?origin=scopusAI

16. AI - Powered Personalised Learning System Balakrishnan, D., Tejasri, Y., Akhila, A., (...), Vaishnavi, D. Proceedings of the 9th International Conference on Electronics, Communication and Aerospace Technology, ICECA 2025, 2025 https://www.scopus.com/pages/publications/105034833576?origin=scopusAI

17. Artificial Intelligence in E-Learning: Shaping Personalized Education through Adaptive Systems Sahani, C., Rawat, N., Kumar, R., Kandpal, V. 2025 International Conference on Computational Engineering, Sensing Technology and Management, ICCETM 2025, 2025 https://www.scopus.com/pages/publications/105043731473?origin=scopusAI

18. A Survey on Phishing Attack Taxonomy, Detection Techniques, Datasets, and Security Measures Kaur, K., Jain, A.K. Journal of Applied Security Research, 2025 https://www.scopus.com/pages/publications/105007721946?origin=scopusAI

19. Across the Spectrum In-Depth Review AI-Based Models for Phishing Detection Ahmad, S., Zaman, M., Sami Al-Shamayleh, A., (...), Akhunzada, A. IEEE Open Journal of the Communications Society, 2025 https://www.scopus.com/pages/publications/105003099172?origin=scopusAI

20. Intelligent Cyber Threat Prediction Using Machine Learning for Phishing and SQL Injection Sajja, S.L., Spoorthy, M., Varsha, M., (...), Sharmila, M. CCIC 2026 - Contemporary Computing Innovations Conference 2026, 2026 https://www.scopus.com/pages/publications/105038781237?origin=scopusAI

21. Susceptibility to instant messaging phishing attacks: does systematic information processing differ between genders? Lee, Y.Y., Gan, C.L., Liew, T.W. Crime Prevention and Community Safety, 2023 https://www.scopus.com/pages/publications/85151501050?origin=scopusAI

22. Which Phish Is on the Hook? Phishing Vulnerability for Older Versus Younger Adults Sarno, D.M., Lewis, J.E., Bohil, C.J., Neider, M.B. Human Factors, 2020 https://www.scopus.com/pages/publications/85068267112?origin=scopusAI

23. Exploring Neurodivergent Learners Remote Learning Experience: Preliminary Results for Students Experiencing ADHD Soror, A., Mahfouz, D., Hassan, A. 29th Annual Americas Conference on Information Systems, AMCIS 2023, 2023 https://www.scopus.com/pages/publications/85192914531?origin=scopusAI

24. SafeSurf-AI: A Browser-Integrated Hybrid Phishing and Spam Detection Framework with Real-Time Awareness Simulation and Explainable AI Kirupa, V., Bhushan, P., Mithun, P. 2025 IEEE 3rd Global Conference on Wireless Computing and Networking, GCWCN 2025, 2025 https://www.scopus.com/pages/publications/105037321790?origin=scopusAI

25. Enhancing Phishing Resilience in Academia: The Mediating Role of Anti-Phishing Tools on Student Awareness and Behavior Alqahtani, S., Nanda, P. 2024 17th International Conference on Security of Information and Networks, SIN 2024, 2024 https://www.scopus.com/pages/publications/86000011469?origin=scopusAI

26. CyberShield - Phishing Detection Browser Plugin: A Machine Learning Approach to Web Security Manu, L., Jayanthi, R., Gond, C.M. 2026 4th International Conference on Artificial Intelligence and Machine Learning Applications: Healthcare and Internet of Things, AIMLA 2026, 2026 https://www.scopus.com/pages/publications/105041611829?origin=scopusAI

27. SecureNet Lite: An AI-Powered Cybersecurity Platform for Smart Web Threat Detection Nishanth, R., Bharadwaj, T.G., Prakash, S., Surya Rangan, H.P. ICKECS 2026 - Proceedings of 4th IEEE International Conference on Knowledge Engineering and Communication Systems, 2026 https://www.scopus.com/pages/publications/105041777306?origin=scopusAI

28. Demo: LLM-Based Browser Extension for Phishing Detection Pham, E., Bui, T., Chen, H. 2025 Silicon Valley Cybersecurity Conference, SVCC 2025, 2025 https://www.scopus.com/pages/publications/105016172537?origin=scopusAI

29. Reviewing Cybersecurity Awareness Training Tools Used to Address Phishing Attack at the Workplace Alghenaim, M.F., Bakar, N.A.A., Rahim, F.B.A. Information Sciences Letters, 2022 https://www.scopus.com/pages/publications/85127483847?origin=scopusAI

30. A Gamified Approach to Enhancing Cybersecurity Awareness: Educating Users on Threats and Best Practices Aswini, J., Mohan, N.A.P., Aravindhnath, T.R., (...), Sathish, R. 2025 IEEE International Conference on Advanced Computing Technologies, ICACT 2025, 2025 https://www.scopus.com/pages/publications/105033618325?origin=scopusAI

31. Using an Interactive Online Quiz to Recalibrate College Students' Attitudes and Behavioral Intentions About Phishing Perrault, E.K. Journal of Educational Computing Research, 2018 https://www.scopus.com/pages/publications/85036612003?origin=scopusAI

32. AI-Based Phishing Detection and Student Cybersecurity Awareness in the Digital Age Shahbazi, Z., Jalali, R., Molaeevand, M. Big Data and Cognitive Computing, 2025 https://www.scopus.com/pages/publications/105014457200?origin=scopusAI

33. Cybersecurity Awareness: Investigating Students' Susceptibility to Phishing Attacks for Sustainable Safe Email Usage in Academic Environment (A Case Study of a Nigerian Leading University) Okokpujie, K., Kennedy, C.G., Nnodu, K., Noma-Osagha, E. International Journal of Sustainable Development and Planning, 2023 https://www.scopus.com/pages/publications/85149211377?origin=scopusAI

34. Assessing user's susceptibility and awareness of cybersecurity threats Althobaiti, M.M. Intelligent Automation and Soft Computing, 2021 https://www.scopus.com/pages/publications/85105062068?origin=scopusAI

35. Enhanced Phishing Detection using Machine Learning, NLP and Emerging Technologies Jeevitha, D., Venkadesh, P., Divya, S.V., (...), Hariharan, S. 2025 8th International Conference on Circuit, Power and Computing Technologies, ICCPCT 2025, 2025 https://www.scopus.com/pages/publications/105020192444?origin=scopusAI

36. Enhancing Web Security: An Efficient URL Phishing Classifier based on Deep Learning Abhijna, S., Vinay Prasad, M.S. 15th International Conference on Advances in Computing, Control, and Telecommunication Technologies, ACT 2024, 2024 https://www.scopus.com/pages/publications/85208811501?origin=scopusAI

37. Advanced techniques and best practices for phishing detection Kumar, R., Gupta, B.B., Psannis, K., (...), Khullar, T. Critical Phishing Defense Strategies and Digital Asset Protection, 2025 https://www.scopus.com/pages/publications/86000108532?origin=scopusAI

38. Preparation and Analysis of Information Security Events for Deep Learning Models in University Networks Sobol, A., Kochyn, V. Nonlinear Phenomena in Complex Systems, 2026 https://www.scopus.com/pages/publications/105036574430?origin=scopusAI

39. Efficiency of an educational game for teaching the behavior of distinguishing legitimate emails from phishing attempts Fontoura, J.M., Henklain, M.H.O., Lobo, F.L., Feitosa, E.L. Journal of the Brazilian Computer Society, 2026 https://www.scopus.com/pages/publications/105040208249?origin=scopusAI

40. Reinforcing Cybersecurity Awareness through Simulated Phishing Attacks: Findings from an HEI Case Study Ciupe, A., Orza, B. IEEE Global Engineering Education Conference, EDUCON, 2024 https://www.scopus.com/pages/publications/85199083426?origin=scopusAI

41. Real Time Cybersecurity Threat Monitoring & Classification System Ganesh, D., Sunil Kumar, M., Varalakshmi, K., (...), Ashok, K. Proceedings of 4th International Conference on Electronics and Renewable Systems, ICEARS 2026, 2026 https://www.scopus.com/pages/publications/105035723414?origin=scopusAI

42. Cyber threat analysis using natural language processing: Automation and profiling Anne, S., Ranga, R., Natte, A., (...), Rao, K.G. AIP Conference Proceedings, 2026 https://www.scopus.com/pages/publications/105042164043?origin=scopusAI

43. College Students' Use and Perceptions of Artificial Intelligence (AI): A Survey Study Strayhorn, T.L. Journal of College Student Development, 2025 https://www.scopus.com/pages/publications/105008876721?origin=scopusAI

44. Widely used but barely trusted: understanding student perceptions on the use of generative AI in higher education Cabrera, C., Neville, R. Perspectives: Policy and Practice in Higher Education, 2025 https://www.scopus.com/pages/publications/105025059551?origin=scopusAI

45. AI use and student perceptions of reliability and academic integrity Zohourian, M., Pamidimukkala, A., Kermanshachi, S. Discover Education, 2026 https://www.scopus.com/pages/publications/105052043979?origin=scopusAI

46. Exploring the use of generative artificial intelligence tools among University Students in Ghana Kwakye, I.N., Kwakye, K.A.P., Adom-Fynn, D., (...), Agyei, S.D. Discover Education, 2025 https://www.scopus.com/pages/publications/105019374365?origin=scopusAI

47. Gender Dynamics of Cyber Security Awareness in Kerala, India: Statistical and Machine learning Approach Swapna, M.P., Rajeev, D., Ramkumar, J., Chandran, S. 2025 Gender and Technology Conference, GTC 2025, 2025 https://www.scopus.com/pages/publications/105037630430?origin=scopusAI

48. Adoption of artificial intelligence tools among pharmacy students in Syria: patterns of use, educational perceptions, and institutional barriers Alajlani, M., Alnokkari, A., Aljerf, L. BMC Medical Education, 2026 https://www.scopus.com/pages/publications/105048275349?origin=scopusAI

49. Artificial Intelligence Tool Adoption in Higher Education: A Structural Equation Modeling Approach to Understanding Impact Factors among Economics Students Sova, R., Tudor, C., Tartavulea, C.V., Dieaconescu, R.I. Electronics (Switzerland), 2024 https://www.scopus.com/pages/publications/85205064670?origin=scopusAI

50. Prediction of the Use of Generative Artificial Intelligence Through ChatGPT Among Costa Rican University Students: A PLS Model Based on UTAUT2 Cabero-Almenara, J., Palacios-Rodríguez, A., Rojas Guzmán, H.D.L.Á., Fernández-Scagliusi, V. Applied Sciences (Switzerland), 2025 https://www.scopus.com/pages/publications/105001054548?origin=scopusAI

51. Uncovering adoption personas for generative AI in higher education: a clustering-based segmentation approach Saihi, A., Ahmed, V. International Journal of Educational Technology in Higher Education, 2026 https://www.scopus.com/pages/publications/105030965794?origin=scopusAI

52. Hybrid Deep Learning-Powered Secure Browser Extension for Real-Time Malicious Website Detection Yogesh, Daniel, E., Neeba, T., (...), Durga, S. International Conference on NexGen Networks and Cybernetics, IC2NC 2025 - Proceedings, 2025 https://www.scopus.com/pages/publications/105035625249?

origin=scopusAI

53. Predicting The Likelihood of Phishing Attacks using Machine Learning Kumar, N.R.S., Vigneshkumar, M., Reddy, A.S.K., (...), Raza, A. Proceedings of 5th International Conference on Communication, Computing and Electronics Systems, ICCCES 2026, 2026 https://www.scopus.com/pages/publications/105037453653?origin=scopusAI

54. SafeClick - A Cloud Native Solution for Phishing Website Detection Deepika, S., Sai, S.S., Sai, P.H., (...), Anitha, C. CCIC 2026 - Contemporary Computing Innovations Conference 2026, 2026 https://www.scopus.com/pages/publications/105038725717?origin=scopusAI

55. Retrieval-Augmented Generation: A Survey of Security Challenges and Countermeasures Wang, C., Li, H., Song, W., Lin, Y. Proceedings - 2025 11th IEEE International Conference on Privacy Computing and Data Security, PCDS 2025, 2025 https://www.scopus.com/pages/publications/105018920964?origin=scopusAI

56. Enhancing Retrieval-Augmented Generation through Modular Optimization: Evaluation Across the RAG Pipeline Mattos, E., Amoedo, D., Guimãraes, V., (...), Waldir Júnior, S.S. 2025 International Symposium on Internet of Things, SIoT 2025, 2025 https://www.scopus.com/pages/publications/105033891186?origin=scopusAI

57. Trusted RAG: Enhancing Business Document QA Accuracy with Hybrid Retrieval, Reranking, and Faithfulness Verification Shreya, Deepika Proceedings of the 3rd International Conference on Innovations in Cybersecurity and Data Science, ICICDS 2026, 2026 https://www.scopus.com/pages/publications/105046108209?origin=scopusAI

58. Enhancing Communication and Data Transmission Security in RAG Using Large Language Models Gummadi, V., Udayaraju, P., Sarabu, V.R., (...), Venkataramana, S. 4th International Conference on Sustainable Expert Systems, ICSES 2024 - Proceedings, 2024 https://www.scopus.com/pages/publications/85214782970?origin=scopusAI

59. User-Centric Phishing Detection: A RAG and LLM-Based Approach Al Barwani, A.H., Korba, A.A., Anwar, R.W. Digest of Technical Papers - IEEE International Conference on Consumer Electronics, 2026 https://www.scopus.com/pages/publications/105037357088?origin=scopusAI

60. Hybrid Email Phishing Detection Using Large Language Models and Bayes Classifiers Chiosa, C., Pungila, C. Communications in Computer and Information Science, 2026 https://www.scopus.com/pages/publications/105039331181?origin=scopusAI

61. CLASP: Cost-Optimized LLM-based Agentic System for Phishing Detection Trad, F., Chehab, A. International Conference on Electrical, Computer, and Energy Technologies, ICECET 2025, 2025 https://www.scopus.com/pages/publications/105037119955?origin=scopusAI

62. Phishing Detection in the Gen-AI Era: Quantized LLMs vs Classical Models Thapa, J., Chahal, G., Gabreanu, S.V., Otoum, Y. Proceedings - 29th IEEE/ACIS International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing, SNPD 2025-Summer, 2025 https://www.scopus.com/pages/publications/105031093724?origin=scopusAI

63. Lightweight URL-based phishing detection using natural language processing transformers for mobile devices Haynes, K., Shirazi, H., Ray, I. Procedia Computer Science, 2021 https://www.scopus.com/pages/publications/85115259702?origin=scopusAI

64. Artificial Neural Networks with Soft Attention: Natural Language Processing for Phishing Email Detection Optimized with Modified Metaheuristics Lakicevic, B., Spalevic, Z., Volas, I., (...), Bacanin, N. Communications in Computer and Information Science, 2025 https://www.scopus.com/pages/publications/105000897444?origin=scopusAI

65. Detecting Human-Written and AI-Generated Phishing Emails with DistilBERT and Explainable AI Kumar, A.A., Imthias Ahamed, T.P., Shukoor, N.A. 2025 2nd IEEE International Conference for Women in Computing, InCoWoCo 2025, 2025 https://www.scopus.com/pages/publications/105036333999?origin=scopusAI

66. Deploying Distilled AI Models for Email Phishing Detection Jahan, N., Imran, M.R., Jahan, M.M., Sazzad, T.M.S. 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence and Networking, QPAIN 2026, 2026 https://www.scopus.com/pages/publications/105042777686?origin=scopusAI

67. Multi-Agent Phishing Detection And Deletion via Small VLM and LLM Reasoning Ayrin, F.J., Rahman, M., Rahman, A., (...), Reza, M.T. 2026 5th International Conference on Electrical, Computer and Telecommunication Engineering, ICECTE 2026, 2026 https://www.scopus.com/pages/publications/105036969482?origin=scopusAI

68. Towards Explainable and Educational Phishing Detection: A Zero-Shot LLM Approach Zhang, L., Wu, Y. Proceedings of the IEEE International Conference on Big Data, BigData, 2025 https://www.scopus.com/pages/publications/105037875453?origin=scopusAI

69. Effects of Personal Characteristics on Phishing Awareness, Anti-Phishing Tool Usage, and Phishing Avoidance Behavior: A Structural Equation Modeling Approach Alqahtani, S., Nanda, P. 2024 17th International Conference on Security of Information and Networks, SIN 2024, 2024 https://www.scopus.com/pages/publications/86000010967?origin=scopusAI

70. Can I live? College student perceptions of risks, security, and privacy in online spaces Haltinner, K., Sarathchandra, D., Lichtenberg, N. Communications in Computer and Information Science, 2016 https://www.scopus.com/pages/publications/84955267883?origin=scopusAI

71. Identifying the level of user awareness and factors on phishing attempt among students Nagalingam, V., Samy, G.N., Ahmad, R., (...), Ibrahim, R. Advanced Science Letters, 2015 https://www.scopus.com/pages/publications/84960433507?origin=scopusAI

72. Cybercrime awareness and digital self-protection among Austrian secondary school students Unterweger, M., Hörmann, C., Kuka, L., Sabitzer, B. Discover Global Society, 2026 https://www.scopus.com/pages/publications/105045494266?origin=scopusAI

73. Neurodivergent Students' Views on Internet Threats: A Call for Research Stone, B., Saggers, B., Campbell, M., Yates, C. Students' Online Risk Behaviors: Psychoeducational Predictors, Outcomes, and Prevention, 2025 https://www.scopus.com/pages/publications/105007393063?origin=scopusAI

74. Asian new generation's perceptions regarding network fraud Jaw, G.-Y., Chen, J.-Y. Second International Conference on Innovative Computing, Information and Control, ICICIC 2007, 2007 https://www.scopus.com/pages/publications/39049124715?origin=scopusAI

75. Modeling students' email security behavior toward phishing threats using PMT for gamified cybersecurity awareness design Gwenhure, A.K., Umnouyvittayakul, A., Ord, L.L., (...), Nam, S. Information and Computer Security, 2026 https://www.scopus.com/pages/publications/105050673417?origin=scopusAI

76. What influences digital natives' susceptibility to cyber phishing scams? Ashrafi, D.M., Sattar, M., Jalal, R. Journal of Financial Crime, 2025 https://www.scopus.com/pages/publications/105018685407?origin=scopusAI

77. Will the "Phisher-men" reel you in? assessing individual differences in a phishing detection task Welk, A.K., Hong, K.W., Zielinska, O.A., (...), Mayhorn, C.B. International Journal of Cyber Behavior, Psychology and Learning, 2015 https://www.scopus.com/pages/publications/84958968579?origin=scopusAI

78. Examining the Experiences of Neurodivergent Learners in STEM Fields in Their Transition to and Engagement with Online Learning Bauer, A.J., Boyer, D.M. ASEE Annual Conference and Exposition, Conference Proceedings, 2025 https://www.scopus.com/pages/publications/105038726651?origin=scopusAI

79. Designing a Just-in-Time (JIT), Microlearning, and Gamification Intervention for Phishing Awareness in Micro, Small, and Medium Enterprises (MSMEs) S, M.N.D.U., Mohd, M. Communications in Computer and Information Science, 2027 https://www.scopus.com/pages/publications/105048242343?origin=scopusAI

80. Exposing generational and gender gap in phishing awareness among young adults: A survey experiment Pratama, A.R., Vadila, N., Firmansyah, F.M. AIP Conference Proceedings, 2023 https://www.scopus.com/pages/publications/85151497795?origin=scopusAI

81. CAPTURE THE FLAG AS A LEARNING TOOL TO IMPROVE CYBERSECURITY EDUCATION Schafeitel-Tähtinen, T., Lazarov, W. SEFI 2025 - 53rd Annual Conference of the European Society for Engineering Education: Engineering and Society, Proceedings, 2025 https://www.scopus.com/pages/publications/105031619482?origin=scopusAI

82. Enhanced Feature Extraction for Phishing URL Detection: A Comprehensive Analysis of Structural, Host-Based, Content and N-Gram Attributes Khedekar, L., Pawar, S. Lecture Notes in Networks and Systems, 2026 https://www.scopus.com/pages/publications/105027725662?origin=scopusAI

83. A Novel Approach for Phishing Website Detection using Deep Learning Models Vyshnavi, C., Raj, C.M., Gopakumar, G., (...), Palaniswamy, S. 8th IEEE International Conference on Computational System and Information Technology for Sustainable Solutions, CSITSS 2024, 2024 https://www.scopus.com/pages/publications/85216923483?origin=scopusAI

84. STRENGTHENING CYBERSECURITY EDUCATION FOR UNIVERSITY STUDENTS: BRIDGING VULNERABILITIES AND PROMOTING PROACTIVE DIGITAL SAFETY PRACTICES Ferhataj, A., Memaj, F., Sahatcija, R., Ora, A. Millenium: Journal of Education, Technologies, and Health, 2025 https://www.scopus.com/pages/publications/105012374385?origin=scopusAI

85. Phishing Detection with Intelligent URL Classification based Ensemble Learning Saini, H.K., Kaur, N. Proceedings of the 7th International Conference on Innovative Data Communication Technologies and Application, ICIDCA 2025, 2025 https://www.scopus.com/pages/publications/105031401643?origin=scopusAI

86. PhishGuard: A Hybrid Ensemble Framework for Detecting Malicious URLs Mathew, J.T., Nair, L.S., Menon, H.P., Asmi, P. Proceedings of 8th International Conference on Computing Methodologies and Communication, ICCMC 2025, 2025 https://www.scopus.com/pages/publications/105016905899?origin=scopusAI

87. Towards detection of phishing websites on client-side using machine learning based approach Jain, A.K., Gupta, B.B. Telecommunication Systems, 2018 https://www.scopus.com/pages/publications/85039076014?origin=scopusAI

88. Combining Text and Visual Features to Improve the Identification of Cloned Webpages for Early Phishing Detection Van Dooremaal, B., Burda, P., Allodi, L., Zannone, N. ACM International Conference Proceeding Series, 2021 https://www.scopus.com/pages/publications/85113199820?origin=scopusAI

89. Battling Phish: An Empirical Comparison of ML, DL, and LLM in Phishing URL Detection Across Datasets Champa, A.I., Rabbi, M.F., Zibran, M. Proceedings of the ACM Symposium on Applied Computing, 2026 https://www.scopus.com/pages/publications/105042880195?origin=scopusAI

90. CAUSE: A Causal GNN Framework with Adversarial Co-Evolution for Robust Phishing Detection Ersedo, T.H., Li, J. 2026 6th International Symposium on Computer Technology and Information Science, ISCTIS 2026, 2026 https://www.scopus.com/pages/publications/105044600444?origin=scopusAI

91. NeuroSymbolicPhishDefend for adaptive multimodal phishing detection against evolving artificial intelligence driven threats Lakshmi, G., Swetha, P. Scientific Reports, 2026 https://www.scopus.com/pages/publications/105049637758?origin=scopusAI

92. Adversarial Robustness of Phishing Email Detection Models Gholampour, P.M., Verma, R.M. IWSPA 2023 - Proceedings of the 9th ACM International Workshop on Security and Privacy Analytics, 2023 https://www.scopus.com/pages/publications/85159123101?origin=scopusAI

93. SiamesePhish: An Advanced AI-powered URL Embedding Approach with Synthetic URLs for Phishing Detection Carthy, A., Sadman, N., Zulkernine, F. Lecture Notes in Computer Science, 2026 https://www.scopus.com/pages/publications/105046071222?origin=scopusAI

94. Proposal of a Model Based on Deep Learning Techniques for the Detection of Phishing in Social Networks Sotomayor, J., García, N., Ticona, W. Lecture Notes in Networks and Systems, 2025 https://www.scopus.com/pages/publications/105014354755?origin=scopusAI

95. Methods, Obstacles, and Potential Use Cases of Using Natural Language Processing to Detect Cybercrimes Yadav, G., Joshi, N.V., Mishra, S.S., (...), Tyagi, S. 2025 World Conference on Cutting-Edge Science and Technology, WCCEST 2025, 2025 https://www.scopus.com/pages/publications/105036592570?origin=scopusAI

96. LightPhishAI: A Lightweight Multimodal Feature Extraction and Fusion Framework for Real-Time Phishing Detection on IoT Devices Lakshmi, G., Swetha, P. SSRG International Journal of Electronics and Communication Engineering, 2026 https://www.scopus.com/pages/publications/105034415140?origin=scopusAI

97. Intelligent cybersecurity systems for phishing attack detection - An overview Vennela, A., Akarapu, R.B., Rakshith, B.L., (...), Sunil, G. Computers and Electrical Engineering, 2026 https://www.scopus.com/pages/publications/105022500100?origin=scopusAI

98. Secure and Ethically Aligned AI Solutions for Cybersecurity and Threat Detection Kolhar, A., Sridevi Ethical AI Applications for Software Engineering and Development, 2026 https://www.scopus.com/pages/publications/105035395137?origin=scopusAI

99. AI and Machine Learning Applications in Ethical Hacking: Advancing Cybersecurity Defense Subramanian, R.S., Kaneti, V.R., Saranya, G., Ponnuviji, N.P. Ethical Hacking and Penetration Testing in Cybersecurity, 2026 https://www.scopus.com/pages/publications/105049496421?origin=scopusAI

100. Leveraging Artificial Intelligence and Machine Learning Techniques for Proactive Threat Detection, Anomaly Prediction, and Adaptive Defense Mechanisms in Cybersecurity Frameworks Patidar, G., Pandey, A., Safdari, A., (...), Ranjan, N. Proceeding - 2026 IEEE 6th International Conference on Computing, Power, and Communication Technologies, IC2PCT 2026, 2026 https://www.scopus.com/pages/publications/105045255125?origin=scopusAI

101. Artificial intelligence and machine learning in security Odeh, A., Alhajahjeh, T., Taleb, A.A. Intelligent and Secure Solutions for Digital Transformation, 2025 https://www.scopus.com/pages/publications/105029507708?origin=scopusAI

102. Cyber Threat Intelligence Analytics: Leveraging Machine Learning Rekha, C., Usha, J. 2025 2nd International Conference on New Frontiers in Communication, Automation, Management and Security, ICCAMS 2025, 2025 https://www.scopus.com/pages/publications/105030053894?origin=scopusAI

103. FedPhishLLM: A privacy-preserving and explainable phishing detection mechanism using federated learning and LLMs Li, W., Manickam, S., Chong, Y.-W. Journal of King Saud University - Computer and Information Sciences, 2025 https://www.scopus.com/pages/publications/105018871117?origin=scopusAI

104. LLM-Driven Data Augmentation with Contrastive Learning to Address Mobile Phishing Attacks: A Persuasion-Theory Approach Shim, H.S., Park, H., Lee, K., Kim, D. Proceedings - 2025 IEEE 45th International Conference on Distributed Computing Systems Workshops, ICDCSW 2025, 2025 https://www.scopus.com/pages/publications/105030458793?origin=scopusAI

105. Virtual Data Augmentation: A Robust and General Framework for Fine-tuning Pre-trained Models Zhou, K., Zhao, W.X., Wang, S., (...), Wen, J.-R. EMNLP 2021 - 2021 Conference on Empirical Methods in Natural Language Processing, Proceedings, 2021 https://www.scopus.com/pages/publications/85123447960?origin=scopusAI

106. Explainable Artificial Intelligence Framework for Real-Time Threat Detection in Smart Cybersecurity Systems Jayalakshmi, R., Pavanathil, R.J., Swetha, C.V., (...), Srinivasan, R. 2026 ASU International Conference in Emerging Technologies for Sustainability and Intelligent Systems, ICETSIS 2026, 2026 https://www.scopus.com/pages/publications/105042759135?origin=scopusAI

107. Explainable artificial intelligence in threat detection Wadhwa, K., Babbar, H., Rani, S. Explainable Artificial Intelligence (XAI) for Next Generation Cybersecurity: Concepts, Challenges and Applications, 2025 https://www.scopus.com/pages/publications/105026745476?origin=scopusAI

108. XAI-Driven Security Systems: Enhancing Trust and Clarity in Automated Threat Detection and Response Maram, B., Balija, S., Sekhar, M.C., (...), Sathishkumar, V.E. 2nd International Conference on Intelligent Systems for Cybersecurity, ISCS 2025, 2025 https://www.scopus.com/pages/publications/105034865505?origin=scopusAI

109. AI Explainability and Trust in Cybersecurity Operations Bhatia, A., Kumar, A. Deep Learning Innovations for Securing Critical Infrastructures, 2025 https://www.scopus.com/pages/publications/105005136429?origin=scopusAI

110. Responsible AI on National Cybersecurity Strategy with Enhancement of Cybersecurity in Education Tingare, B.A., Prasanna Lakshmi, G., Khedkar, V., (...), Khatkale, P.B. International Conference on Intelligent and Innovative Practices in Engineering and Management 2024, IIPEM 2024, 2024 https://www.scopus.com/pages/publications/105001921359?origin=scopusAI

111. Auditing AI-Generated cyber threats in intelligent systems: A risk-based framework with legal considerations Al-Zoubi, T.M.H. EDPACS, 2026 https://www.scopus.com/pages/publications/105035413880?origin=scopusAI

112. Ethical AI Frameworks For Governance Of Intelligent Systems Chelliah, E.A.A., Pathak, G., Tripathi, S., (...), Bhavadharani, S. International Journal of Artificial Intelligence and Machine Learning, 2026 https://www.scopus.com/pages/publications/105044527184?origin=scopusAI

113. VerifyWise - A Modular AI Governance Platform for Data-Driven Compliance Management Khalilzadeh, M., Razavi, F. 2026 International Interdisciplinary Conference on Artificial Intelligence: Engineering, Health, Finance and Humanities, IICAI 2026, 2026 https://www.scopus.com/pages/publications/105047083905?origin=scopusAI

114. A Comprehensive Mentoring Model for Cybersecurity Education Wang, P., Sbeit, R. Advances in Intelligent Systems and Computing, 2020 https://www.scopus.com/pages/publications/85085735401?origin=scopusAI

115. Integrating deep learning techniques for personalized learning pathways in higher education Naseer, F., Khan, M.N., Tahir, M., (...), Aejaz, S.M.H. Heliyon, 2024 https://www.scopus.com/pages/publications/85195324322?origin=scopusAI

116. Design of Structure using an Adaptive Generative AI System for Customised Eduction for Individuals Sharma, S., Raj, M., Ali, M.M., (...), Singh, S. Proceedings - International Conference on Computing, Power, and Communication Technologies, IC2PCT 2024, 2024 https://www.scopus.com/pages/publications/85191163306?origin=scopusAI

117. APPLICATION OF MOBILE LEARNING BASED ON ARTIFICIAL INTELLIGENCE IN STUDENT OPEN TEACHING STRATEGY Zhang, X., Wang, P., Guo, W. Transactions of the Royal Institution of Naval Architects Part A: International Journal of Maritime Engineering, 2025 https://www.scopus.com/pages/publications/105019062007?origin=scopusAI

118. Federated Learning Driven Privacy-Preserving Cybersecurity for Sustainable Cross-Organization Threat and Fraud Detection Simi, S., Aswathy, R.H., Sheeja Kumari, V., Wilfred Blessing, N.R. Proceedings of 9th International Conference on Trends in Electronics and Informatics, ICOEI 2026, 2026 https://www.scopus.com/pages/publications/105042346044?origin=scopusAI

119. Phishing email detection using MBFT-Lite: A multi-branch federated transformer framework Shrotriya, A., Agarwal, S., Khanna, T., Singhal, S. Journal of Discrete Mathematical Sciences and Cryptography, 2026 https://www.scopus.com/pages/publications/105033637525?origin=scopusAI

120. A Survey on Federated Fine-Tuning of Large Language Models Wu, Y., Tian, C., Li, J., (...), Xu, C. Transactions on Machine Learning Research, 2026 https://www.scopus.com/pages/publications/105030625083?origin=scopusAI

121. A Survey on Federated Parameter-Efficient Fine-Tuning for Large Language Models Wen, Q., Zhang, X., Xiang, N., (...), Zhang, J. International Conference on Big Data and Information Analytics, BigDIA, 2025 https://www.scopus.com/pages/publications/105034880400?origin=scopusAI

122. Federated Learning with Explainable AI (XAI) for Natural Language Processing in Large-Scale Distributed Systems Immidisetti, V.R., Akula, R.P., Kumar, M., (...), Singh, G. International Conference on Engineering, Technology and Management, ICETM 2026, 2026 https://www.scopus.com/pages/publications/105048496403?origin=scopusAI

123. Enhancing Business-Specific Phishing Chat Detection via Few-Shot Learning LLM Augmentation Hansakul, S., Wisitpongphan, N. IEEE Access, 2025 https://www.scopus.com/pages/publications/105019690191?origin=scopusAI

124. Memory-Aware Privacy Protection for Large Language Models in Education Liao, Y., Keung, J., Chan, W.-K. Proceedings - 2025 International Symposium on Educational Technology, ISET 2025, 2025 https://www.scopus.com/pages/publications/105015825333?origin=scopusAI

125. Differentially private data augmentation via LLM generation with discriminative and distribution-aligned filtering Song, Y., Zhang, J., Tian, Z., (...), Li, D. Neural Networks, 2026 https://www.scopus.com/pages/publications/105029669234?origin=scopusAI

126. Retrieval-Augmented Generation for Secure Environments: Theoretical Foundations, Agent Architectures, and Local Deployment with FAIRD Gaida, E., Grund, N., Ventulett, N., (...), Conrad, J. Lecture Notes in Computer Science, 2026 https://www.scopus.com/pages/publications/105028360080?origin=scopusAI